

Cover Letter

Candidate: Ahmet Halit Ünsal

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Position: Head of Embedded Software Engineering

Target Organization: AVL Turkey (Istanbul, Turkey)

Job URL: <https://www.avl.com/careers/head-of-embedded-istanbul>

Dear Hiring Team & Leadership at AVL Turkey,

I am writing to express my strong interest in the Head of Embedded Software Engineering position at AVL Turkey.

With over 15 years of professional engineering experience - including 8+ years leading and directing multi-disciplinary engineering organizations of up to 30 engineers across 4 team leads - my background directly aligns with AVL Turkey's requirements for high-reliability embedded software architecture, Model-Based Design (MBD), and mission-critical powertrain controls.

Throughout my leadership tenure across automotive and defense electronics programs (NISO Technology, ECEMTAG, TÜBİTAK, TÜMOSAN), I have directed full product lifecycles from initial concept and dynamic modeling to dyno characterization and mass production. I maintain strict enforcement of ISO 26262 ASIL D functional safety, ASPICE Level 2/3 processes, UN R155/R156 cybersecurity (CSMS), and AUTOSAR layered architectures across FreeRTOS and bare-metal platforms.

On the technical front, I bring hands-on mastery in MATLAB/Simulink/Stateflow physical modeling, PMSM/IPMSM Field-Oriented Control (FOC, MTPA, flux-weakening), and complex EV powertrain controls (VCU, MCU, Traction Inverters, OBC, BMS). Complementing this is extensive experience establishing automated Hardware-in-the-Loop (HIL) simulation testbenches (dSpace MABX, Lauterbach Trace32) and Git-based CI/CD pipelines that dramatically compress verification cycles and eliminate integration bottlenecks.

Beyond embedded systems, my end-to-end technical leadership is demonstrated in platforms like EduTrace (edutrace.net), highlighting my ability to scale complex architectures and bridge software, hardware, and product teams. I would welcome the opportunity to discuss how my technical ownership, executive leadership, and delivery focus can accelerate the engineering milestones at AVL Turkey.

Sincerely,

Ahmet Halit Ünsal

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